

Evaluation of the impact of a pharmaceutical care program in children with asthma

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Abstract

The objective of this study was to evaluate the impact of a pharmaceutical care program on children with asthma. A comprehensive asthma education and monitoring program that includes basic asthma knowledge, symptoms and exacerbation evaluation, pharmacotherapy assessment including inhaler technique, and quality of life measurements was developed and applied in an outpatient paediatric clinic of the Catholic University of Chile. All patients with moderate asthma scheduled for outpatient visits with their internist over a 1-year period were referred for pharmacist intervention. Patients (aged 7–17) with moderate asthma attending the clinic were allocated to the intervention (group A) or control group (group B). Intervention patients were educated on their disease, pharmacotherapy, self-management, and inhalation techniques. The group B were children with their regular treatment for asthma but without pharmaceutical intervention. A paediatric asthma quality of life questionnaire (PAQLQ) was applied to both groups at 0, 2, and 9 weeks to assess the quality of life. Spirometry was done at the beginning and at the completion of the 9-week study. Beta-agonists used by each patient were also recorded. Eleven children (10.0 ± 0.7 years) were included in the pharmaceutical care program, and ten children (9.9 ± 0.6 years) in group B. For the individual domains of activities (A), emotions (E), and symptoms (S) there was a significant improvement in the children who received pharmaceutical care in comparison with those who did not receive it. The scores of group B did not change during the 9 weeks of follow-up. There were no significant changes in spirometric values in either group.

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1. Introduction

Traditionally, clinicians and clinical investigators have used conventional measurement of asthma, such as airway caliber spirometry [1], symptoms [2], use of beta-agonists [3], and responsiveness of the airway to exercise and cold air [4], as components of comprehensive assessment of the patient's status. However none of these parameters indicates whether the patients are actually able to function better in their day-to-day lives. It has been assumed that if one or more of these indices has improved then the patient's health-related quality of life must be improved. This may not always be the case. Children with asthma are troubled not only by symptoms such as shortness of breath, cough, and wheeze, but they are also bothered by the physical, social, educational, and emotional impairments that they experi-

ence as a result of having asthma. More sensitive measures are needed to capture subtle changes in asthma management [5]. Consequently, disease-specific quality-of-life measures have been developed and show promise for asthma research. Juniper et al. [6] developed a questionnaire called the asthma quality of life questionnaire, designed to evaluate quality of life in children with asthma between 7 and 17 years. This instrument has been evaluated and validated by these authors [7–9].

The role of pharmacists in patient education and evaluation of therapy is taking on more relevance every day. Their knowledge of drug therapy makes them the appropriate professionals to instruct and educate the patient on the correct use of medications. For many years, pharmacists have been taking an active role not only in the education of the patient, but also in the monitoring and evaluation of therapy, and detecting potential problems (adverse reactions and interactions, therapy failures, non adherence to treatment, duplication of therapy, etc.) that directly affect the quality of life of patients [10–12]. Pharmacists are in an ideal

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position to assess and treat adherence-related problems that can adversely affect patients' health outcomes. Pharmaceutical care is a new practice paradigm that has evolved from many years of research and practice in the profession of pharmacy. This new professional practice is not intended to replace the role of the physician or any other practitioner but rather to meet a need in the health care system that has arisen because of multiple prescribes of medication for a single patient, the explosion in the number of drug products on the market, the increased complexity of drug therapy, and the significant level of drug-related morbidity associated with the drug use [13]. Strategies to monitor and improve adherence are key components of pharmaceutical care plans, particularly for patients with chronic diseases, such as chronic lung disease [14–18], arterial hypertension [19–22], and diabetes mellitus [23,24]. The present work attempts to measure the impact of a pharmaceutical care program in asthmatic children by comparing treatment alone versus treatment with pharmaceutical care.

2. Materials and methods

2.1. Patients

We recruited children with stable and moderate asthma, as defined by the American Thoracic Society guidelines [25,26], who were scheduled for outpatient visits with their internist over a 1-year period at the outpatient Clinic of the Department of Pediatrics of the Catholic University of Chile. Patients were eligible if they were between the ages of 7 and 17 years, had moderate asthma, and if they required medications daily.

We excluded children who had illnesses other than asthma that affected quality of life. Children over 12 years and all of the caregivers signed an informed consent that had been approved by the Ethics Committee of the Faculty of Medicine.

2.2. Interventions

In this study we used a pediatric asthma quality of life questionnaire (PAQLQ), Spanish version [27] to measure the impact of a pharmaceutical care program in two groups of children: the intervention group (group A), children with moderate asthma who received pharmaceutical care during 9 weeks, and control group (group B), children with moderate asthma who did not receive pharmaceutical care during this time. Both groups were receiving medications for their condition.

Dr. Juniper from the Department of Clinical Epidemiology and Biostatistics, MacMaster University Medical Center, Hamilton, Ont., Canada provided the PAQLQ Spanish version [27]. The PAQLQ is a disease-specific quality of life questionnaire that has been developed to measure the physical, emotional, and social impairments that are experienced by children (7–17 years of age) with asthma. It has

been rigorously tested in all age groups (7–9, 10–13, and 14–17 years) to ensure that it is reproducible (repetition in stable subjects produces more or less the same results), that it is valid (it actually measures quality of life in asthma), and that it is responsive to change (able to detect important changes in quality of life, even if those changes are small) [7]. This questionnaire contains 23 items in three domains: activities (five items), asthma symptoms (10 items), and emotional function (eight items). Scores on each item can range from 1 to 7 where 1 indicates maximum impairment and 7 indicates no impairment. At admission to the program, a clinical pharmacist, trained to apply the questionnaire, met with the child and administered the PAQLQ. In the activity domain, three of the items are “individualized”. At the first visit, each patient was asked to identify three activities that were done on a regular, frequent basis and that were troublesome because of asthma. These individualized activities were retained for each patient throughout the study (e.g. playing with friends, riding a bicycle, running).

Afterwards, the child was assigned at random to one of the two groups of study: group A or group B. Both written and verbal instructions were given to parents and children from group A. In addition to the prescriptions and the educational information, in a face-to-face session, patients also received an explanatory booklet (prepared especially for this study) about antecedents of the illness, the alternatives of treatment, and what to do in case of an asthmatic crisis. They also received a second information brochure about every medication prescribed by the physician, explaining in simple terminology the mode of action of these drugs, and the adverse effects and problems that could present in the course of the therapy. At this stage, parents and children were instructed about the correct use of the inhaler for drug administration. Each session lasted about 30 min. After this, the pharmacist and children with their parents scheduled a visit for 2 weeks later.

Children in group B were only administered the PAQLQ without receiving any special instruction. They received only their prescribed medication and the usual recommendations of their physician. At the second and third follow-up visits, both groups were administered the PAQLQ again. Children were interviewed and made their ratings without their parents being present. In addition to the questionnaire, parents and the children included in group A were given reinforcement of the information provided in the first visit. The pharmacist also asked the parents about concerns in relation to the medications, and any adverse drug reactions observed during treatment. All the children included in the study were evaluated in three follow-up visits immediately after entering the program, and 15 days and 9 weeks after first receiving the pharmacist's instructions and drug treatment.

For each patient, a spirometry measurement was made at the beginning of the therapy and following completion of the 9 weeks of follow-up. Both groups were educated equally in the technique of spirometry.

2.3. Data analysis

Individual items within the PAQLQ are equally weighted. The questionnaire is analysed directly from the scores recorded and the results are expressed as the mean score per item for each of the domains. We considered patients whose score was 1, 2, or 3 as having experienced an unimportant change. Those with scores of 4 or 5 were considered to have experienced a moderate change, and those with scores of 6 or 7 were considered to have experienced a large change.

We used a two-factor ANOVA with repeated measures to evaluate treatment differences, the effect of time, and the interaction between treatment and time. In order to compare the groups, and their spirometric values, we used the two sample Student's *t*-test. Values of $P < 0.05$ were considered significant.

3. Results

3.1. Clinical characteristics of the children included in the study

All 21 recruited children completed the study. Thirteen children were female and eight were male. Eleven children were included in group A (six male and five female) and ten in group B (seven male and three female). A summary of clinical characteristics of patients enrolled in the study is given in Table 1. The mean age $\pm$ S.D. of all patients in the study was 9.9 ± 0.4 years (range 7–14 years). The mean age of group A and B $\pm$ S.D. was 10.0 ± 0.6 and 9.9 ± 0.6 years respectively. At the beginning of the study, both groups had similar forced expiratory volume in 1 s (FEV₁), 2.4 ± 0.8 l (S.D.) (group A) and 2.3 ± 0.2 l (group B). This difference was not statistically significant.

3.2. Medication

Table 2 shows the drugs prescribed by the physician at the beginning of study. Both groups used the same type of medications in a similar regimen. Albuterol inhalation (up to 200 μ g q.i.d.) was the rescue drug, mostly used by the children in both groups. Inhaled beclomethasone (50–200 μ g b.i.d.), fluticasone (250–500 μ g b.i.d.), or budesonide (200–400 μ g q.d.) were the drugs most often prescribed.

Table 1
Clinical characteristics of the patients included in the study

Patients	Group A	Group B	<i>P</i>
Number	11	10	
Age years ($\pm$ S.D.)	10.0 ± 0.6	9.9 ± 0.6	N.S.
Sex F/M	5/6	3/7	
FEV ₁ (L) ($\pm$ S.D.)	2.4 ± 0.8	2.3 ± 0.2	N.S.

Note: FEV₁: Forced expiratory volume in one second.

Table 2

Drugs and number of patients whom were receiving the drug at the beginning of the study

Inhaled drugs	Group A (number of patients)	Group B (number of patients)
Beclomethasone	7	6
Fluticasone	3	2
Budesonide	1	2
Albuterol (prn)	10	10

Note: prn: as needed by the patient.

Table 3

Baseline score for each domain measured by quality of life questionnaire (mean $\pm$ S.D.)

Domains	Group A	Group B	<i>P</i>
Activities	3.8 ± 0.3	4.0 ± 0.3	N.S.
Symptoms	4.1 ± 0.5	4.6 ± 0.4	N.S.
Emotions	5.2 ± 0.4	5.2 ± 0.4	N.S.

3.3. Quality of life questionnaire

Table 3 shows the baseline scores after applying the questionnaire to both groups of children. There were no statistically significant differences between the groups.

3.3.1. Emotions domain

Fig. 1 shows the curves for the emotions domain in groups A and B. At the first 2 weeks point of the study, differences were not statistically significant between groups. However, at the 9th week, group A showed an statistically significant improvement in the score. In group B, who did not receive

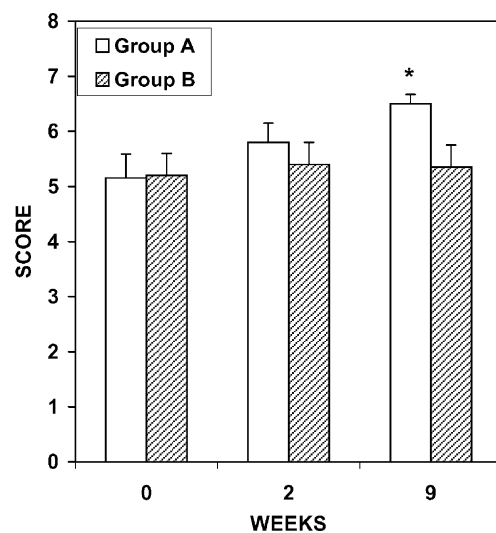


Fig. 1. Changes in emotions score in group A ($N = 11$) and group B ($N = 10$) after 9 weeks of study. The values for emotions score after 2 and 9 weeks are compared with the corresponding value in week 0; * $P < 0.01$ at week 9.

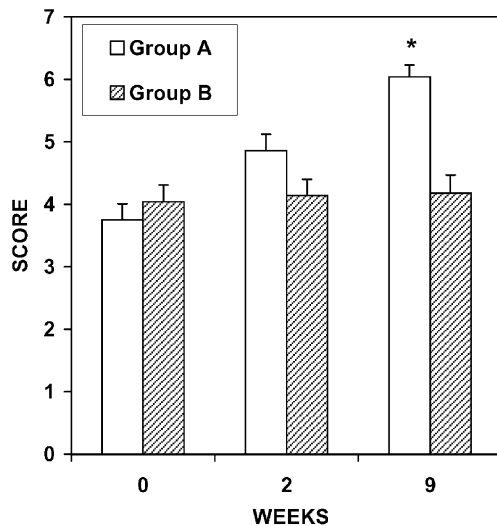


Fig. 2. Changes in activities limitations score in group A ($N = 11$) and group B ($N = 10$). The values for activities limitations score after 2 and 9 weeks are compared with the corresponding value in week 0; * $P < 0.001$ at week 9.

pharmacist counseling, the emotions scores were maintained at approximately the same level during the entire 9 weeks of treatment.

A two-factor ANOVA with repeated measures was calculated comparing the scores between treatments and the time. The score change for treatments was significant ($F(1, 19) = 28.3$, $P < 0.0001$). There was significant score change for time ($F(2, 38) = 38.3$, $P < 0.0001$). The interaction between groups and time was also statistically significant ($F(2, 38) = 20.67$, $P < 0.0001$).

3.3.2. Activities limitations domain

Fig. 2 shows the changes observed in the children of group A in physical activities scores. Similarly to the emotions domain, there was an improvement at the 9th week of the study. group B did not have any change in their physical activities scores.

The score change for treatments was significant ($F(1, 19) = 229$, $P < 0.0001$). There was significant score change for time ($F(2, 38) = 210.5$, $P < 0.0001$). The interaction between groups and time was also statistically significant ($F(2, 38) = 151.5$, $P < 0.0001$).

3.3.3. Symptoms domain

Fig. 3 shows the curves for group A and group B in symptoms scores. group A experienced an improvement in their symptoms (fatigue, cough, wheezing, chest tightness, shortness of breath, sleep disturbance, difficulty taking a deep breath) after the 9th weeks of the study. group B did not experience any improvement. The score change for treatments was significant ($F(1, 19) = 14.33$, $P < 0.001$). There was significant score for time ($F(2, 38) = 74.6$, $P < 0.0001$). The interaction between groups and time was also statistically significant ($F(2, 38) = 50$, $P < 0.0001$).

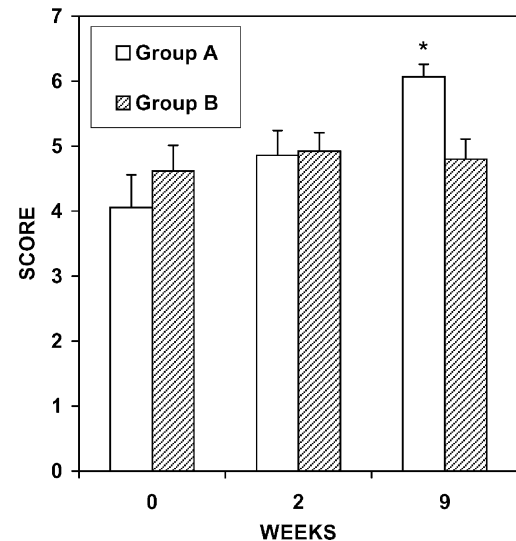


Fig. 3. Changes in symptoms score in group A ($N = 11$) and group B ($N = 10$) after 9 weeks of study. The values for symptoms score after 2 and 9 weeks are compared with the corresponding value in week 0; * $P < 0.002$ at week 9.

Table 4

Spirometry measurements $\pm$ S.D. in the group A at the beginning and after 9 weeks of the study, in comparison with the group B

Group	FVC	FEV ₁	FEF 25–75	FEF 50
Group A				
Before	3.08 $\pm$ 0.97	2.41 $\pm$ 0.76	2.28 $\pm$ 0.86	2.75 $\pm$ 1.08
After	3.13 $\pm$ 1.14	2.48 $\pm$ 0.89	2.30 $\pm$ 0.77	2.65 $\pm$ 0.91
Group B				
Before	2.66 $\pm$ 0.19	2.34 $\pm$ 0.22	2.82 $\pm$ 0.63	3.17 $\pm$ 0.63
After	2.85 $\pm$ 0.29	2.51 $\pm$ 0.27	3.10 $\pm$ 0.59	3.51 $\pm$ 0.72

Note: FVC: Forced vital capacity; FEV₁: Forced expiratory volume in one second; FEF 25–75: Forced expiratory flow during 25–75% FVC; FEF50: Forced expiratory flow during 50% FVC.

3.3.4. Spirometry

Table 4 shows the results of the spirometry at the beginning and end of the study. No statistical differences in the parameters were seen between the groups or within the groups.

4. Discussion

The provision of individualized written information to patients, increases the accuracy of their recall of details about their illness and its treatment [28]. In asthma in particular, randomized controlled trials have shown that written patient information booklets can reduce acute contacts for medical care [29]. In the present study, the provision of pharmaceutical care to children with asthma significantly improved their quality of life. It has also been demonstrated that the questionnaire designed by Juniper et al. [7] was effective for assessing the improvement of emotions,

physical activity, and symptoms. The observed changes for each one of the studied domains were statistically significant in the 9th week after providing patient counseling by the pharmacist. These changes could be explained because the pharmaceutical care was not limited to information in booklets or oral advice, but the pharmacist also identified any drug therapy problems that might be interfering with the goals of the therapy. Specifically the pharmacist detected subjective concerns of the parents about the use of corticosteroids. Some parents thought that these medications could retard the child's growth and they refused to administer it. The pharmacist explained to parents that corticosteroids administered by inhaler are not absorbed from the lungs and for this reason, the fraction of drug that reach the blood is not enough to produce systemic effects. In many or most instances, simply providing information about the disease or recommended behaviour changes is insufficient for changing health-related behaviour. The pharmacist in this study also gave recommendations about the correct use of inhalers to the parents and children and reinforced the indications given by the physician. The education and the interventions provided to the parents and children seem to be important for the success of treatment, due to the fact that children and their parents often ignore or fail to remember instructions given by the physician. This statement was confirmed in our study because patients in group B, which did not receive counseling by the pharmacist, did not improve their quality of life during the time of the study.

The spirometric values showed that there were no significant differences between baseline values in both groups. The fact that we did not find significant changes after the study, could be due to several reasons: first, the asthma was mild in severity, with normal spirometry or mild obstructive pattern. Therefore, there was a narrow possible range to improve in the measurements that we evaluated. Second, the study may not have been sufficiently long to evaluate changes in pulmonary function tests [30]. On the other hand, variability in pulmonary function may have had an influence because the sample size was small. Finally, there is no close correlation between quality of life, symptoms, and pulmonary function tests [31]. We postulate that over a longer study period, we might be able to demonstrate a significant effect on pulmonary function. Aside from the clinical implications of our data, the study provided interesting insights into parents' attitudes about asthma and its drug treatment. Parents had many doubts and myths about the disease and the medications that were clarified by the pharmacist.

5. Conclusion and practice implications

The present study demonstrated that the pharmaceutical care provision to children with asthma and their parents significantly improved the children's quality of life and improved knowledge about the disease and its treatment.

The visits that children and their parents made to the pharmacist were very useful in clarifying many myths and mistaken concepts about drug therapy and the disease. All these pharmacist interventions were reflected in an improvement in the quality of life of the children.

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